FAQ-11: How Do Metrics Work in Dynatrace?

> **Series:** FAQ — Frequently Asked Questions | **Reference:** 11 — How Metrics Work in Dynatrace | **Created:** July 2026 | **Last Updated:** 10/23/2026

Overview

Every observability conversation eventually lands on a metric — a CPU line on a dashboard, an error-rate alert, an SLO burn rate. But *what a metric actually is* in Dynatrace, and what happens to it between the moment a measurement is taken and the moment a chart renders, is rarely spelled out in one place. The answer determines things teams usually discover the hard way: why a query over a year of data comes back in seconds, why a `builtin:` metric key returns nothing in a Notebook, why an OpenTelemetry counter silently never arrives, why one careless dimension can consume a cardinality budget, and why metric queries do not show up on the DPS bill.

This FAQ is decision support for the question: **what happens to a metric in Dynatrace end to end — how a data point is produced, the shape it is normalized into, where it is stored (and for how long, at what resolution), how it is queried, and what it costs — and where that model constrains your design choices?**

It goes deep on the mechanics: the data-point model (gauge summaries and count deltas), the two coexisting metric backends (Metrics Classic and Metrics powered by Grail) and the dual-write bridge between them, every documented ingest path (OneAgent, the ingest APIs, OTLP, Extensions 2.0, and metric extraction from records), the storage and rollup model, dimension-cardinality enforcement, the DQL `timeseries` query path, and the DPS billing rules. It complements **FAQ-09** (*When Should I Query a Metric Instead of Raw Logs?*), which covers *when* to reach for a metric; this entry covers *how metrics themselves work*.

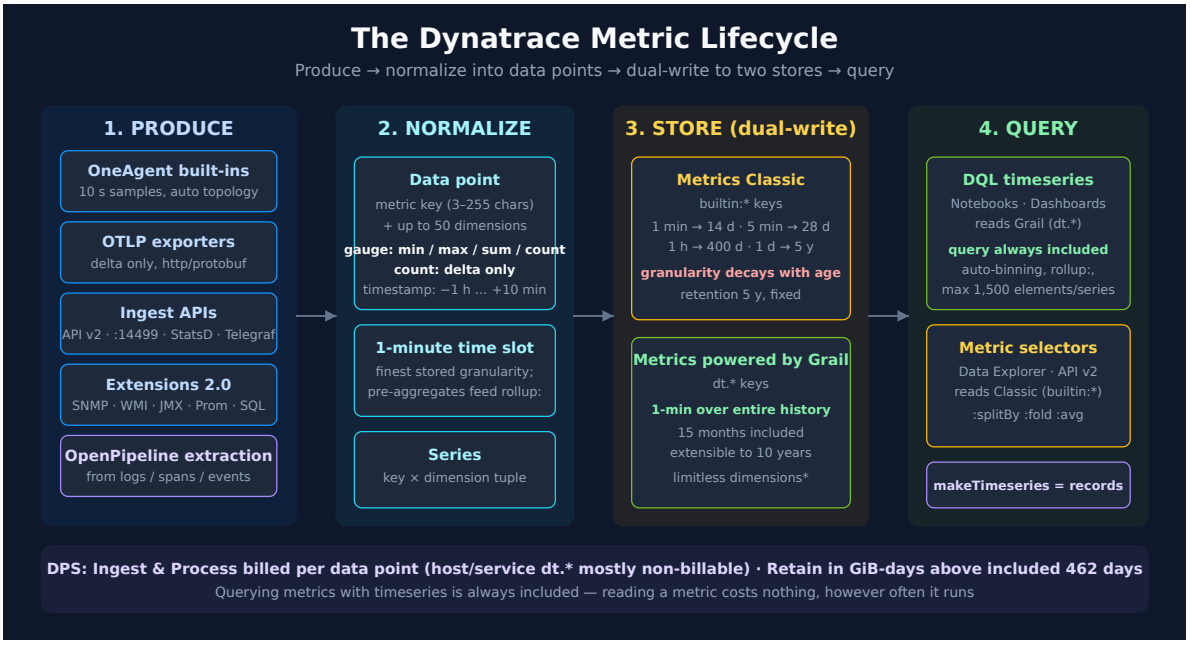


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Prerequisites

Requirement	Details
Audience	Platform engineers, SREs, observability architects, and anyone designing custom metrics, OTLP pipelines, or dashboards — and FinOps owners who want to understand what drives metric cost
Format	Decision-support deep dive — explains the metric model end to end; not a hands-on lab
Deployment	Dynatrace SaaS. The Grail-specific sections assume Grail/DPS; the Metrics Classic sections apply to classic-surface assets (Data Explorer, Dashboards Classic, Metrics API v2)
Related topic series	OTEL (OpenTelemetry ingest), OPIPE-03/04 (metric extraction from spans, cardinality management), OPLOGS-03 (metric extraction from logs), FINOPS-01/02/03 (DPS consumption querying, forecasting, optimization), DASH (dashboarding), SLO (metrics behind SLOs)
Related FAQs	FAQ-09 (when to query a metric instead of raw logs — the economics companion to this entry)

1. Short Answer

A Dynatrace metric is a **pre-aggregated numeric time series**: a metric key, a set of dimensions (up to 50), and a stream of timestamped data points that Dynatrace aggregates into **one-minute time slots** at ingest. A gauge data point is not a single number — it carries a `min` / `max` / `sum` / `count` summary of everything measured in its slot, which is why a query can later ask for the average, the peak, or the total of the *same* stored point without re-reading raw data. Counters are stored as **deltas**, never as cumulative totals.

That one design decision — *aggregate once at write time, read the aggregate forever* — explains most metric behavior you will encounter:

Layer	What happens	Why it matters
Produce	OneAgent samples infrastructure every 10 seconds; apps push via OTLP or the ingest APIs; extensions poll SNMP/JMX/Prometheus/SQL/WMI; OpenPipeline extracts metrics from logs, spans, and events	Many producers, one model — everything converges on the same line-protocol data-point shape
Normalize	Every measurement becomes a gauge summary (`min` , `max` , `sum` , `count`) or a count delta on a metric key + dimensions	The stored pre-aggregates are what DQL's `rollup:` parameter later selects among
Store	Dual-write into two separate databases: Metrics Classic (granularity decays with age) and Metrics powered by Grail (1-minute granularity over the entire retention window)	Which backend a tool reads determines what resolution and history it sees
Query	DQL `timeseries` (Notebooks, Dashboards) reads Grail; metric selectors (Data Explorer, Metrics API v2) read Classic	Classic `builtin:` keys are invisible to DQL — you query the Grail-native `dt.*` twin instead

Layer	What happens	Why it matters
Bill	DPS bills metric **Ingest & Process** per data point and **Retain** (in GiB-days) only above the included 15 months; querying metrics with `timeseries` is *always included*	Reading a metric ten thousand times costs the same as reading it once: nothing

If you remember one thing: **metrics are shaped at ingest, not at query time.** Dimensions, units, temporality, and cardinality are all decisions the *producer* makes; the query side can only slice what was stored. The rest of this FAQ walks each layer in depth.

> **Sources:**
>
> - Metric ingestion protocol (DT docs)
> - Metric API FAQ (DT docs) — "The finest granularity of a time slot is one minute."
> - Metrics FAQ (DT docs)
> - Metrics powered by Grail capability (DT docs) — "Querying metrics using the `timeseries` command is always included."
> - **Derived:** the five-layer framing is an authoring synthesis of the cited pages; the docs describe each layer separately

2. The Metric Data Model

2.1 The metric key

A metric key is a dot-separated path (`myapp.orders.processed`). The ingestion protocol defines the rules verbatim: allowed characters are *"lowercase and uppercase letters, numbers, hyphens (`-`), and underscores (`_`)"*; *"metric keys cannot start with a number or a hyphen"*; sections cannot start with a hyphen; non-Latin letters are not allowed; and *"the length of the key must be in range from 3 to 255 characters."*

The **prefix carries meaning** — it tells you who produced the metric and which backend generation it belongs to:

Prefix	Who writes it	Notes
`dt.*`	Dynatrace (Grail-native built-ins)	**Reserved** — custom keys starting with `dt.` are rejected at ingest
`builtin:*`	Dynatrace (Classic built-ins)	The Classic twin of `dt.*` keys; queryable via metric selectors, not DQL
`ext:*`	Extensions (Classic naming)	Extension-framework metrics
`calc:*`	Calculated metrics (Classic)	e.g., calculated service metrics; `calc:apps.*` RUM metrics are not supported on Grail
`log.*`	Log metric extraction	Metrics based on a log monitoring query carry the `log.` prefix
(your namespace)	You	Free-form — in community practice, prefix with an org or app namespace so ownership stays obvious

For Grail-native keys, Dynatrace's own naming convention is instructive: the docs describe *"replacing the `builtin:` prefix with `dt.` to clearly denote Dynatrace-provided metrics"* and *"preferring snake case (`capacity_units`) to camel case (`capacityUnits`)"*.

2.2 Dimensions

Dimensions are key-value pairs attached to a data point — *"you can specify up to 50 dimensions"* per metric line. Dimension keys allow lowercase letters, numbers, hyphens, periods, colons, and underscores; values are quoted strings (quotes and backslashes escaped). If the same dimension key appears twice on one line, only one value is accepted.

The critical concept dimensions create is the **series**: one metric key × one unique dimension-value tuple = one series, and each series produces its own stream of data points. Cardinality (§7) and billing (§9) are both counted in series, not in keys.

You can see the series structure of your own tenant with the DQL `metrics` discovery command — this query (validated live) counts series per host-metric key; the per-disk metrics unsurprisingly carry the most:

```
metrics from: now()-24h
| filter startsWith(metric.key, "dt.host.")
| summarize series = count(), by: {metric.key}
| sort series desc
| limit 10
```

2.3 Data points: gauges and counts

Every data point is one of two payload types:

- **Gauge** — a sampled state ("CPU is at 80.6%"). A single value `gauge,80.6` is expanded at ingest to the summary `min=80.6, max=80.6, sum=80.6, count=1`; producers that pre-aggregate can send the summary form directly (`gauge,min=17.1,max=17.3,sum=34.4,count=2`). Those four stored statistics are exactly what DQL's `rollup:` parameter selects among at query time (§8).
- **Count** — an occurrence tally ("500 requests since the last report"). The protocol is explicit: *"data points of the `count` type are deltas between the previous and current data points"* — you send `count,delta=500`, never a cumulative running total. On **Metrics Classic**, a count key is automatically suffixed with `.count` (and a gauge key that already ends in `.count` gets `.gauge`); **Grail does not do this** — *"Grail metrics do not apply the .count and .gauge suffixes."* In DQL, query the key exactly as sent.

Timestamps are UTC milliseconds, and the acceptance window is tight: *"between 1 hour into the past and 10 minutes into the future from now"* — anything outside is rejected. Omit the timestamp and the server clock is used.

2.4 Metadata: units, display names, descriptions

Metadata makes a metric render meaningfully (a chart labeled "MilliSecond" instead of a bare number). Two mechanisms:

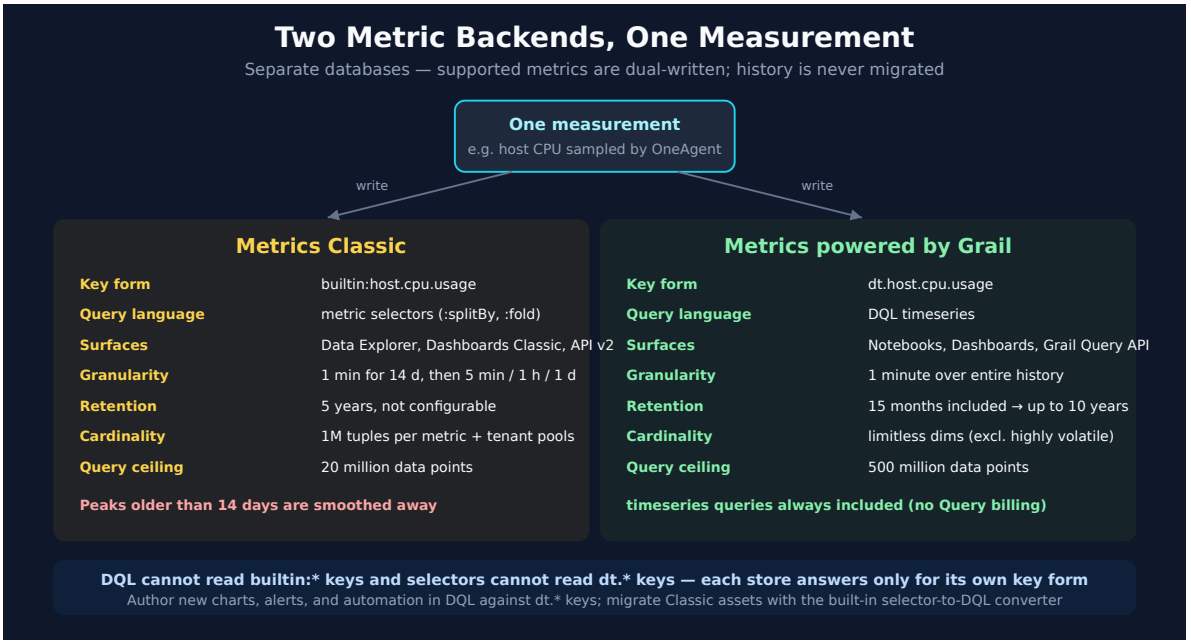
1. **At ingest**, the line protocol supports a metadata line: `#my.metric gauge dt.meta.displayName="...", dt.meta.description="...", dt.meta.unit="..."`.
2. **After ingest**, the Settings schema `builtin:metric.metadata` (scope `metric-{your-metric-key}`) manages display name, description, unit, tags, dimension display names, and Davis-relevant properties (`rootCauseRelevant`, `impactRelevant`, `minValue` / `maxValue`).

One documented limitation to plan around: *"You cannot provide metadata for built-in or calculated metrics; metadata is supported only for custom ingested metrics"* — built-in metadata is registered by Dynatrace itself.

> **Sources:**
>
> - Metric ingestion protocol (DT docs) — key/dimension rules, gauge summary expansion, count deltas, timestamp window, and the Classic-only `.count` / `.gauge` suffixing (*"Grail metrics do not apply the .count and .gauge suffixes."*)
> - Metric limits (DT docs) — `dt.` prefix reserved for Dynatrace
> - Built-in metrics on Grail (DT docs)
> - Built-in classic metrics (DT docs) — `builtin:` / `ext:` / `calc:` prefix semantics
> - Custom metric metadata (DT docs) — metadata mechanisms and the built-in/calculated limitation

3. Two Backends: Metrics Classic vs Metrics Powered by Grail

Dynatrace currently operates **two metric databases side by side**, and understanding the split resolves most "why can't I see my metric here?" confusion. The docs are unambiguous: *"Classic metrics and Grail are separate databases with separate storage."* Nothing is migrated between them — instead, *"supported metrics are already being written to both Metrics Classic and Grail"* (dual-write), so each backend accumulates its own history from the point support began.



3.1 The key mapping

Built-in metrics exist under **two names**: the Classic key (`builtin:host.cpu.usage`) and the Grail-native key (`dt.host.cpu.usage`). These are the same measurement dual-written to two stores — but each store only answers for its own key form. This is the single most common trip-wire when teams move to Notebooks:

- **DQL cannot query `builtin:` keys.** A `timeseries` on a `builtin:` key either fails to parse (the colon reads as parameter syntax) or, backtick-quoted, returns nothing — the Grail store has no such key. Query the `dt.*` twin instead. We confirmed this live: on a current tenant, the DQL `metrics` command lists hundreds of `dt.*` keys and zero `builtin:` keys.
- **Metric selectors cannot query `dt.*` keys** — they read Classic.

The Grail catalog is also consolidated, not 1:1. The built-in-metrics page states that Dynatrace *"modified the availability of service metrics in Grail, which might affect the direct relation between Metrics Classic and Grail metric keys"*, and frames the change as dimensional rather than additive — endpoint request metrics are *"produced automatically for every detected endpoint"*, with out-of-the-box dimensions replacing calculated metrics for common splits. In practice that is the `dt.service.*` family split by dimensions instead of key proliferation; the upgrade guide puts it as *"Over 100 Classic service metrics are consolidated into a handful of Grail-based metrics, accessed through dimensions and filters."* Some Classic families have no Grail twin — *"calculated RUM metrics (metrics with the prefix `calc:apps`) are not supported on Grail"* and security metrics were replaced by security events.

3.2 Which tool reads which backend

Surface	Backend	Query language
Notebooks, Dashboards (Latest Dynatrace)	Grail	DQL `timeseries`
Grail Query API (`/platform/storage/query/...`)	Grail	DQL
Data Explorer, Dashboards Classic	Classic	Metric selector
Metrics API v2 (`/api/v2/metrics/query`)	Classic	Metric selector

The migration state, per the docs: *"most but not all metrics are already supported on Grail"* — service, infrastructure, cloud, container/Kubernetes, and runtime families are available, with others not yet supported — the metrics FAQ marks the remaining gaps, such as custom buckets for Grail metrics and deletion of Grail metric data, as *"planned for a future release."* SaaS 1.349 (pre-release; rollout planned from 10/06/2026) announces the first of them: *"Metrics now supports custom buckets."* Verify it has reached your tenant; until then the current guidance remains the working path. Data Explorer and Dashboards Classic ship a built-in **metric-selector-to-DQL converter** to help move assets. It does not cover everything: for what it cannot handle, the

upgrade guide says to *"manually convert them by mapping Classic aggregations, transformations, and filters to their DQL equivalents."* Classic's "Auto" pseudo-aggregation is the common snag — *"As DQL doesn't have an equivalent pseudo-aggregation, you need to specify the aggregation explicitly."*

Practical rule: author every *new* chart, alert, and automation in DQL against `dt.*` keys; treat metric selectors as the maintenance language for existing Classic assets until you migrate them (§8.4).

> **Sources:**
>
> - Metrics FAQ (DT docs) — separate databases, dual-write, tool-to-backend mapping
> - Metrics on Grail upgrade guide (DT docs) — migration state, service-metric consolidation
> - Built-in metrics on Grail (DT docs) — `builtin:` ↔ `dt.` key mapping tables
> - Metric limits (DT docs)
> - Metric selector conversion (DT docs)
> - Data Explorer (DT docs)
> - What's new in Dynatrace SaaS 1.349 (DT docs) — *"Metrics now supports custom buckets."* (pre-release, read 10/23/2026)

4. How a Data Point Is Born: The Ingest Paths

Every path below converges on the same data-point model from §2 — the differences are *where* the measurement is taken, *who* authenticates, and *what* gets added automatically.

Path	Runs where	Auth	Distinctive behavior
OneAgent built-ins	On the host	none (agent)	10-second sampling → 1-minute aggregates; automatic topology dimensions
Metrics API v2 ingest	Anywhere → cluster/ActiveGate	platform token / OAuth (`openpipeline:metrics:ingest`) or classic token (`metrics.ingest`)	Line protocol, 1 MB payload
OneAgent local API / `dynatrace_ingest` / Telegraf / StatsD	On the host, `localhost` only	none (local trust) — the local API must be enabled first	Automatic host dimensions; 1,000-line payload
OTLP (`/api/v2/otlp/v1/metrics`)	Anywhere → cluster/ActiveGate	platform token (`openpipeline:metrics:ingest`) or classic token (`metrics.ingest`)	**Delta temporality required for counters and histograms**; HTTP + binary protobuf only
Extensions 2.0	OneAgent or ActiveGate (EEC)	framework	Declarative YAML; polls SNMP/WMI/JMX/Prometheus/SQL/Python
OpenPipeline extraction	At record ingest	pipeline config	Metrics derived from logs/spans/events — §5

4.1 OneAgent built-in metrics

The workhorse. Per the docs: *"infrastructure metrics and other periodic metrics are captured every 10 seconds and stored as 1-minute aggregates"* — the aggregate typically carrying min/max/count/sum/average. Service and web-request metrics are different in kind: *"service and web request metrics do not have a frequency"* — they are derived from the observed transactions themselves (medians and 90th percentiles). Not everything is 10-second: some built-ins report at 1-minute intervals, and a few trail by up to 15 minutes. OneAgent also attaches the topology context (host, process group, entity IDs) as dimensions automatically — which is what makes `by:{dt.smartscape.host}` splits work with zero configuration.

4.2 Metrics API v2 ingest — the line protocol

`POST /api/v2/metrics/ingest` accepts a plain-text body of protocol lines, one data point each. Authenticate with a platform token or OAuth client (`Authorization: Bearer …` , scope `openpipeline:metrics:ingest`) or, where classic access tokens are still available, an `Api-Token` with the `metrics.ingest` scope. *"The payload is limited to 1 MB"* and *"there's no limit on the number of metrics"* within it. A `202` means accepted for background processing; a `400` can be a *partial* failure — valid lines are still ingested.

```
# metadata line (optional, once per key)
#myapp.orders.processed count dt.meta.displayName="Orders processed", dt.meta.unit="Count"

myapp.orders.processed,region="us-east-1",tier="checkout" count,delta=42
myapp.queue.depth,region="us-east-1" gauge,17
myapp.request.duration,region="us-east-1" gauge,min=12.1,max=340.7,sum=1893.4,count=25 1751896800000
```

4.3 The local-host ingest family

When code already runs on a OneAgent-monitored host, you can skip tokens entirely — once the local API is switched on. It ships with OneAgent but is enabled per environment or host: **Settings → Preferences → Extension Execution Controller**, turn on *Enable Extension Execution Controller* and *Enable local HTTP Metric, Log and Event Ingest API* (a host-level setting overrides the environment one). Until then, requests to port 14499 fail. The **local metric API** — `http://localhost:14499/metrics/ingest` (served by OneAgent's Extensions Execution Controller) — *"is available only to local clients and cannot be reached from remote hosts."* Payload cap: 1,000 lines. Its convenience feature is enrichment: *"the host ID and host name context are automatically added to each metric as dimensions."* The same channel backs three integrations:

- **`dynatrace_ingest`** — a CLI shipped in `<install>/agent/tools` ; pipe protocol lines via stdin or pass one per invocation. Ideal for cron jobs and shell scripts.
- **Telegraf** — the upstream Telegraf distribution includes a Dynatrace output plugin; point it at the local endpoint (or a cluster URL + token for remote).
- **StatsD** — OneAgent ships a StatsD daemon listening on UDP `18125` ; an ActiveGate can act as a remote listener on `18126` (disabled by default there). The OneAgent daemon is not available when OneAgent is deployed on Kubernetes (for example with Dynatrace Operator) — there, the docs recommend remote StatsD through an environment ActiveGate.

4.4 OTLP — OpenTelemetry metrics

`POST /api/v2/otlp/v1/metrics` (or via ActiveGate) with `Authorization: Bearer <platform token>` (scope `openpipeline:metrics:ingest`) or, where classic access tokens are still available, `Authorization: Api-Token …` with `metrics.ingest` . Three protocol constraints surprise OTel veterans:

1. **HTTP only** — *"gRPC is not supported"* — and **binary protobuf only** (*"JSON is not supported"*): set `OTEL_EXPORTER_OTLP_PROTOCOL=http/protobuf` .
2. **Counters and histograms must be delta**: *"the Dynatrace backend exclusively works with delta values and requires the respective aggregation temporality."* Cumulative counters and cumulative histograms are not ingested — *"Cumulative histograms aren't ingested (similarly to cumulative counters)."* A cumulative UpDownCounter (a non-monotonic sum) is accepted and stored as a gauge, per the page's instrument mapping table. Fix it at the SDK (`OTEL_EXPORTER_OTLP_METRICS_TEMPORALITY_PREFERENCE=DELTA`) or in a Collector with the `cumulativetodelta` processor (set `max_staleness` above the receive interval so counter state is not evicted between reports).
3. **Histogram support is partial**: explicit-bucket histograms (Dynatrace 1.300+) are stored as histograms **only when Advanced OTLP metric dimensions is enabled** — otherwise they are ingested as counters, with no buckets to take percentiles from; for exponential histograms *"Dynatrace ingests the histogram's min/max/sum/count but doesn't ingest the buckets"*; histogram points without a sum (negative recordings) are dropped; OTLP *summary* metrics are not supported at all.

Attribute mapping: data-point attributes become dimensions by default; resource and scope attributes pass through an allow-list — unless the tenant enables **Advanced OTLP metric dimensions**, in which case *all* resource/scope/data-point attributes become dimensions except a deny list, and `otel.scope.name` / `otel.scope.version` are always added. On conflicts, data-point attributes win over scope, which wins over resource. Limits shift with that mode too: 50 → 100 dimensions, 100 → 350-character

dimension keys, 255 → 2,048-character values; requests cap at 4 MB uncompressed and 15,000 data points; instrument units longer than 63 characters are silently dropped.

4.5 Extensions 2.0

For technology you cannot instrument directly, an extension is a **declarative YAML** (`extension.yaml` in a signed ZIP) that tells the Extensions Execution Controller what to poll: SNMP, WMI, JMX, Prometheus endpoints, SQL queries, or custom Python. It runs *"either your local data sources when run on OneAgent, or remote data sources when run from an ActiveGate"* and emits through the same ingestion protocol as everything else. See the ONBRD and K8S series for deployment patterns.

> **Sources:**
>
> - Metrics Classic (DT docs) — 10-second capture, 1-minute aggregates, service-metric derivation
> - POST ingest data points (DT docs) — endpoint, 1 MB payload, partial-failure semantics; auth: *"Platform Token / OAuth: One of the following scopes is required: openpipeline:metrics:ingest"*, or an `Api-Token` with `metrics.ingest`
> - Metric ingestion protocol (DT docs)
> - OneAgent metric API (DT docs) — port 14499, local-only, 1,000 lines, host enrichment; *"You only need to enable the OneAgent metric API at the environment or host level."*
> - Scripting integration `dynatrace_ingest` (DT docs)
> - Telegraf (DT docs)
> - StatsD (DT docs) — ports 18125/18126; *"OneAgent deployed on Kubernetes, for example using Dynatrace Operator, isn't supported."*
> - OTLP API (DT docs) — endpoints, HTTP/protobuf-only; *"To authenticate, use a platform token with Bearer in the Authorization HTTP header."*
> - OTLP metrics ingest concepts (DT docs) — delta temporality, histogram type mapping (explicit-bucket histogram → Histogram with Advanced OTLP metric dimensions, Counter without; read 10/02/2026); instrument mapping table rows Counter Cumulative → N/A and UpDownCounter Cumulative → Gauge, and *"Cumulative histograms aren't ingested (similarly to cumulative counters)."* (re-read 10/15/2026)
> - OTLP metrics configuration (DT docs)
> - OTLP metrics limitations (DT docs) — size/length limits, 63-character unit cap
> - Collector configuration (DT docs) — `cumulativetodelta` + `max_staleness` guidance
> - Extensions 2.0 concepts (DT docs)
> - Extension data sources (DT docs)

5. Deriving Metrics from Records: OpenPipeline and Classic Extraction

Not every metric starts life as a measurement — many of the most valuable ones are *derived* from records (logs, spans, events) at ingest. This is the "compute once" move from FAQ-09 applied at the pipeline: instead of re-scanning records on every dashboard render, an extraction rule turns matching records into data points as they arrive.

5.1 OpenPipeline metric extraction (current mechanism)

OpenPipeline's processing stage offers three extraction processors — a **counter metric** (*"Returns the number of occurrences of a metric, from the records that match the query."*), a **value metric** (*"Returns the aggregated values of a metric from the records that match the query."*, reading a numeric field), and a **histogram metric** (SaaS 1.343). Extraction is configurable on logs, generic events, SDLC events, security events (new), business events, system events, user events, and user sessions — and *"extracted metrics are sent to Grail only"* (with narrow exceptions for the security-event and span scopes).

Spans get special treatment. Because spans can be sampled, the span scope offers *sampling-aware* counter, value, and histogram metric processors that extrapolate through the sampling ratio — leave the sampling options enabled so a 1-in-100 sampled trace stream still yields approximately correct request counts. Log- and event-derived metrics have no such caveat: those records are not sampled, so extracted metrics are exact. OPIPE-03 covers the span case in depth; OPLOGS-03 covers logs.

Two design rules carry over from FAQ-09 verbatim: **bound your dimensions** (each unique tuple becomes a stored, billed series — §7) and remember that **extraction is forward-only** — a rule created today produces data points from today onward; it does

not backfill history from stored records.

5.2 Log metrics (`log.*` keys)

Log metrics are now the OpenPipeline extraction above, applied to logs: they are created in the Logs pipeline's **Metric extraction** tab (Settings → Process and contextualize → OpenPipeline → Logs), and *"Log metrics are stored in Grail and can be accessed and queried with Dynatrace Query Language."* Each is a **counter** (a count of matching records), a **value** metric (a numeric attribute recorded as a gauge — average, count, maximum, minimum, sum), or a **histogram** metric (a numeric attribute recorded as a histogram — median and the 10th/75th/90th percentiles). The key prefix is fixed: *"All metrics based on a log monitoring query have a metric key with the log. prefix."*

5.3 Calculated service metrics (Classic) — direction of travel

Calculated service metrics are the classic way to derive request-scoped metrics (by URL pattern, request attribute, etc.). Their status: *"OpenPipeline metric extraction now replaces calculated service metrics as the method to create additional service business or technical metrics"*, and *"If you're an existing customer, you can still use them until we deprecate them. If you're a new customer, you won't be able to create calculated service metrics anymore."* No deprecation date is published (checked 09/18/2026).

Existing calculated service metrics reach Grail by cardinality (upgrade guide, rewritten 09/15/2026). One converts automatically when, among other criteria, *"The metric cardinality is less than 500 within any 5-minute window in the past two weeks"*; above that it must be switched on manually; and *"Classic calculated metrics with a cardinality exceeding 2,000 can't be converted to Grail calculated metrics"* — recreate those with OpenPipeline or DQL. Conversion does not double-bill — *"Calculated service metrics continue to consume your license like custom metrics. There are no changes to billing."* — but it can cost more: classic keeps *"at most 100 dimension values"*, while Grail keeps the full cardinality, so *"converting such a metric may increase the number of stored metric data points"*.

> **Histogram extraction (SaaS 1.343, rollout from 07/14/2026):** *"You can now extract histogram metrics from logs or spans using OpenPipeline."* That gives percentile-capable series from record data without shipping raw histograms through OTLP. A tenant still on an earlier version has the counter and value processors only.
>
> **Sources:**
>
> - OpenPipeline processing (DT docs) — counter/value/histogram processors, supported scopes, Grail-only routing
> - Extract metrics from spans (DT docs)
> - Log metrics (DT docs) — counter/value/histogram log metrics, the `log.` prefix, and Grail storage (page updated 08/20/2026, read 10/02/2026)
> - SaaS 1.343 release notes (DT docs) — histogram metric extraction; *"Rollout start on Jul 14, 2026"*
> - Calculated metrics for services (DT docs) — replacement by OpenPipeline, existing vs new customers
> - Upgrade from calculated service metrics to Grail (DT docs) — cardinality-based conversion, billing, top-100 vs full cardinality

6. Storage: Resolution, Rollups, and Retention

6.1 What is physically stored

The finest stored granularity is the **one-minute time slot** (*"the finest granularity of a time slot is one minute"*). For gauges, each slot holds the pre-aggregated summary — min, max, sum, count — not the raw 10-second samples. This is the load-bearing fact behind DQL's `rollup:` parameter: when you write `avg(dt.host.cpu.usage, rollup: max)`, you are choosing *which stored statistic* per slot feeds the aggregation. Nothing is recomputed from raw samples, because raw samples were never stored.

6.2 Metrics Classic: granularity decays with age

Classic storage downsamples as data ages, on a fixed schedule:

Data age	Stored resolution
0–14 days	1 minute
14–28 days	5 minutes
28–400 days	1 hour
400 days–5 years	1 day

Retention is *"5 years, not configurable."* The practical consequence: a Classic query comparing this month against thirteen months ago is comparing 1-minute data against 1-hour data — fine for trends, misleading for peaks (a 2-minute CPU spike simply no longer exists at 1-hour resolution).

6.3 Metrics powered by Grail: full resolution, configurable retention

Grail metrics keep **"1 minute over entire history."** Included retention is *"15 months (462 days) of 1-minute granularity"*, and *"For the default metrics bucket, the available retention period ranges from 1 day to 10 years (3,657 days)."* Retention beyond the included 15 months bills the Retain capability (§9). Retained data stays queryable until the end of the retention period. Note the current scoping: retention is set on the **default metrics bucket**; custom metric buckets are documented as *"planned for a future release"*, so per-metric retention splits are not yet available the way they are for logs.

The Grail design also removed a classic artifact: Classic kept separate pre-aggregated and full-cardinality "detail" variants of some built-ins for performance; Grail *"is already optimized for such high cardinality queries"*, so only the detailed form exists.

> **Sources:**
>
> - Metric API FAQ (DT docs) — one-minute finest slot
> - Data retention periods (DT docs) — classic downsampling tiers, Grail 15-month default
> - Metric limits (DT docs)
> - Metrics Retain capability (DT docs) — 462 days included, 10-year ceiling
> - Metrics FAQ (DT docs) — custom metric buckets planned
> - Built-in metrics on Grail (DT docs) — *"Since the pre-aggregated metric builtin:host.net.bytesRx is no longer necessary, only builtin:host.net.nic.bytesRx is supported by metrics on Grail"*

7. Dimensions and Cardinality

Cardinality is the count of **unique dimension-value tuples** per metric key — and it is the one resource a metric design can exhaust silently. The math is multiplicative: `endpoint (50) × status (10) × region (6)` is 3,000 potential series; add `customer_id (100,000)` and you have 300 million.

7.1 Enforcement on Metrics Classic

Classic enforces hard limits, measured over *"the dimension tuples of the last 30 days (a 30-day sliding window)"*:

- **Per metric:** 1 million dimension tuples.
- **Per tenant:** pool limits across all custom metrics (50 million tuples) and across all built-in metrics — check the metric limits page for the current pool values on your tenant generation.

Exceeding a limit does not stop the metric — it stops *growth*: *"data points having new and as yet unseen dimension tuples are rejected"* while existing series keep flowing. This makes the failure mode subtle: dashboards keep rendering, but new hosts/pods/endpoints never appear.

The docs call out the classic trap directly — **volatile dimensions**: *"short-lived IDs and timestamps are volatile dimensions"*, and a dimension that changes per data point *"will reach the cardinality limits after 1 million seconds online — just 11 days."* Never use request IDs, session IDs, trace IDs, timestamps, or unbounded user IDs as metric dimensions; that detail belongs in logs and spans, which are built for it.

Watch it before it bites: Dynatrace ships self-monitoring metrics for dimension-tuple usage, charted on the **"Metric & Dimension Usage + Rejections"** preset dashboard (dimensions below 10% of a limit are omitted from the detail view).

7.2 Grail posture

Metrics powered by Grail advertises *"limitless dimensions (excluding highly volatile dimensions)"* — the storage engine no longer needs a per-metric tuple cap, but the exclusion in parentheses is doing real work: highly volatile dimensions remain outside the design intent, and *"Dynatrace therefore may restrict or reject metric configurations that don't follow best practices."* The custom-metric-key ceiling (100,000 distinct keys) applies to both backends.

7.3 The cardinality–cost interplay

Dimensions do not bill directly — per the DPS docs, *"metric data points are not billed based on an increase in dimensions, but rather by the increased number of metric data points."* The catch is the second clause: every new tuple is a new series, and every series emits data points per interval. Splitting one metric into 1,000 series multiplies ingested data points by up to 1,000. Cardinality discipline is therefore simultaneously a reliability practice (Classic limits) and a cost practice (DPS ingest) — OPIPE-04 and FINOPS-03 §9 cover remediation patterns.

> **Sources:** Metrics API best practices (DT docs) — limits, sliding window, rejection behavior, volatile-dimension math, self-monitoring dashboard, Metric limits (DT docs) — Grail limitless-dimensions posture, 100,000 custom keys, Metrics Ingest & Process capability (DT docs) — dimensions-vs-data-points billing rule.

8. Querying Metrics

8.1 DQL `timeseries` — the Grail query path

`timeseries` is a *starting* command: it loads, filters, and aggregates metric storage into series in one step. The essentials (all validated live on a tenant):

```
timeseries avg(dt.host.cpu.usage), from:-2h, by:{dt.smartscape.host}
| limit 3
```

Mechanics worth internalizing:

- **Auto-binning.** *"The timeseries command automatically calculates an appropriate time interval derived from the query timeframe"* — snapped to well-known slots (1, 2, 3, 5, 10, 15, 30 minutes; 1–24 hours; multiples of 24 hours up to 30 days), overridable with `interval:` or `bins:`. A series never exceeds the hard cap of **1,500 elements**, but the default lands well below it: on 10/02/2026 a 2-hour query returned 1-minute slots and a 365-day query returned **3-day** slots, 122 elements each. Use `interval:` or `bins:` when you need finer resolution.
- **`rollup:` selects the stored statistic** (§6.1) — `avg`, `min`, `max`, `sum`, `total` — independently of the across-series aggregation. It is **required** for `percentile` / `median` / `percentRank` on gauge and count metrics; omit it and the query returns no rows plus a WARNING notification (`SX_WARNING_TIMESERIES_MISSING_ROLL_UP_TYPE`) — easy to miss if you read only the records. The documented exceptions are `dt.service.request.response_time` and `dt.service.request.service_mesh.response_time`.
- **`scalar: true`** collapses a series to one number over the whole timeframe — cheaper than materializing an array and post-processing it.

```
// p90 is the percentile across all hosts' series in each time slot
timeseries {
    p90 = percentile(dt.host.cpu.usage, 90, rollup: avg),
    avg_cpu = avg(dt.host.cpu.usage, scalar: true)
  }, from:-2h
```

Also available: `filter:` (pre-aggregation dimension filtering), `shift:` (timeframe comparison — day-over-day in one query), `default:` (fill empty slots), and `union:`. Output is normalized: *"all series have identical start and end timestamps, time interval and number of elements."*

8.2 `metrics` — discovery, not charting

The `metrics` command lists **series** (keys + dimension tuples), returning no values or timestamps. Its documented constraints: the timeframe *"is limited to the last ten days"*, at most 100,000 series per query, and it needs `storage:metrics:read` + `storage:buckets:read`. Use it to answer "what exists, with which dimensions?" — as in the §2.2 discovery query.

8.3 `makeTimeseries` — the record-side contrast

`makeTimeseries` builds series **from records at query time** (`fetch logs/events/spans | makeTimeseries …`). Functionally similar output; economically opposite input. In our live validation, metric `timeseries` queries scanned **0 billable bytes**, while this equivalent-shaped event aggregation scanned 110,350 records (0.23 GB) — the record scan bills the Query capability of its data type; the metric read bills nothing:

```
fetch events, from:-2h
| makeTimeseries events = count(), interval: 15m
```

That asymmetry is the entire argument of FAQ-09: recurring aggregations belong in metrics; `makeTimeseries` is for ad-hoc analysis of records you are already scanning.

8.4 Metric selectors — the Classic query path

The Metrics API v2 (`/api/v2/metrics/query`) and Data Explorer consume **metric selectors** — a transformation-chain syntax (`builtin:host.cpu.usage:splitBy("dt.entity.host"):avg:fold(avg)`) with up to 10 metrics per query. It remains a current, non-deprecated API surface, but it reads Metrics Classic only. When migrating assets, the built-in converter maps the common shapes (`:splitBy(...)` → `by:{...}` , `:fold(avg)` → array/scalar aggregation, `median` → `percentile(...,50)` , count-metric `value` → `sum`); Classic's "Auto" aggregation has no DQL equivalent and needs a human decision.

Query-size ceilings differ by an order of magnitude and change what is *feasible*: Classic caps a query at 20 million data points; Grail at **500 million** — high-cardinality, long-window analyses that Classic rejects are routine in Notebooks.

> **Metric metadata via DQL (SaaS 1.343, staged rollout from 07/14/2026).** `load "/dt/platform/metrics.metadata"` returns one row per metric available in the environment — `metric.key` , `name` , `description` , `kind` , `unit` , `dimensions` — as a **separate surface** from the `metrics` command, not an extension of it. Useful for building self-documenting dashboards and for auditing which dimensions a metric actually carries before writing `by:{...}` clauses. On a tenant that has not yet taken 1.343, the `metrics` discovery command above remains the working surface.
>
> `dql > load "/dt/platform/metrics.metadata" > | filter startsWith(metric.key, "dt.host.cpu") > | limit 2 >`
>
> *(Executed 09/28/2026: returned rows with `metric.key` , `name` , `description` , `kind` , `unit` and `dimensions` .)*
>
> **Sources:**
>
> - Metric commands — timeseries, metrics (DT docs) — *"The resulting number of time slots does not exceed the maximum number of elements per series (1,500)."*; *"Two metrics support timeseries percentile without a rollup parameter"*. The binning, `rollup:` warning and percentile examples executed 10/02/2026
> - Aggregation commands — makeTimeseries (DT docs)
> - Metric selector (DT docs) — transformations, 10-metric cap
> - Metrics API v2 (DT docs)
> - Metric selector conversion (DT docs)
> - Metric limits (DT docs) — 20M vs 500M query ceilings
> - SaaS 1.343 release notes (DT docs) — *"You can now query metric metadata via DQL with load"* (rollout from Jul 14, 2026)
> - Metrics powered by Grail capability (DT docs) — timeseries queries always included

> **New in the classic Metrics API (Dynatrace API 1.346, released 08/25/2026): `asHistogram` in the metric descriptor.** The API changelog adds an `asHistogram` enum value to the `transformations` property of the metric descriptor returned by `GET /metrics` and `GET /metrics/{metricKey}` (Environment API v2), and to `metricsMetadata` on the Early-Access `POST /ua/entity` and `POST /ua/list` — so a descriptor can now list `asHistogram` among the transformations a metric supports.
>
> Two things to keep straight. The metric-selector reference does not document an `asHistogram` selector syntax (it documents `:histogram`, which *"Exposes the buckets of a histogram metric as dimensions."*), so verify the syntax before building selector automation on it. And it is a classic-surface change — nothing in the `timeseries` syntax above changes, and DQL still reads Grail only (§3).
>
> **Sources:** Dynatrace API changelog version 1.346 (DT docs) — the `asHistogram` transformations enum addition, re-read 10/02/2026; Metric selector (DT docs) — documents `:histogram`, no `asHistogram`.

9. What Metrics Cost Under DPS

The Metrics powered by Grail capability has three dimensions — **Ingest & Process**, **Retain**, and **Query** — but the third is a formality: *"querying metrics using the `timeseries` command is always included."* Metric reads never appear on the bill. (FAQ-09 reaches the same conclusion from the cost-decision angle: the Query dimension exists but is always included, so metric reads never bill.)

9.1 Ingest & Process — billed per data point

"Ingested data is the number of metric data points sent to Dynatrace", where *"a metric data point is a single measurement of a metric."* The counting rules:

Measurement	Billed data points
Gauge or counter measurement	1
Histogram measurement	10 — *"except `dt.service.*` metric keys which are counted as one data point"*

Scale intuition from the docs' own example: one series reporting every minute is *"1 metric data point × 60 min × 24 h × 365 days = 525,600 metric data points/year."* Multiply by your series count (§7.3) — that, not dimension width, is the bill driver.

9.2 What's included / non-billable

- *"The following metric keys are non-billable: `dt.*`"* — but the exceptions are broad, and they run the other way. **All** `dt.cloud.aws.*`, `dt.cloud.azure.*` and `dt.osservice.*` metrics are billable apart from a short named list (for example `dt.cloud.aws.az.running` and the Azure `vms.*` counts); the NAM `dt.synthetic.multi_protocol.*` metrics are billable; and `dt.service.*` consume the included Full-Stack volume when the spans come from Full-Stack OneAgents and are charged as Metrics powered by Grail otherwise. In short: the host-monitoring catalog is generally free, cloud-integration and OS-service metrics are not; *"some built-in metrics are not billed — this includes, for example, host monitoring metrics included in Full-Stack Monitoring and Infrastructure Monitoring."*
- Three `legacy.*` families — `legacy.containers.*`, `legacy.dotnet.perform.*` and `legacy.tomcat.*` — and everything in the `dt_system_metrics` bucket are non-billable.
- Custom metrics draw first on an included allotment: *"the number of included Custom Metric data points is dependent on the total monitored GiB-hours of your deployment and the OneAgent mode."*
- During a licensing transition there is no double-billing: *"Only the consumption related to your active license applies, whether it's DDUs, DPS Custom Metrics Classic, or DPS Metrics powered by Grail."* (Metrics FAQ (DT docs))

9.3 Retain — billed in GiB-days above the included window

Retain's unit is *"the gibibyte-day (GiB-day)"* — storage volume, not data points — and it only accrues **beyond the included 462 days**. The docs' worked example: at 9 GiB/day of metric data retained for 5 years (1,832 days), billable retention is 9 GiB ×

(1,832 − 462) = 12,330 GiB carried per day. Extending metric retention is cheap relative to record retention, but it is not free — decide it deliberately (§10).

For querying your *own* tenant's metric consumption, FINOPS-01 documents the `dt.system.events` billing-event schema; FINOPS-02/03 cover forecasting and optimization levers.

> **Sources:** Metrics Ingest & Process capability (DT docs) — data-point definition, counting rules, non-billable keys and the billable exceptions (*"The following dt.* metric keys are billable unless otherwise noted in the list above"*, read 10/02/2026; the page lists `legacy.containers.*`, `legacy.dotnet.perform.*` and `legacy.tomcat.*`, and states *"Metrics stored in the bucket dt_system_metrics are non-billable."*), included allotment, Metrics FAQ (DT docs) — no double-billing during a licensing transition, Metrics Retain capability (DT docs) — GiB-day unit, 462 days included, worked example, Metrics powered by Grail capability (DT docs) — capability structure, query always included.

10. Recommended Approach

1. **Inventory before you invent.** Run the `metrics` discovery command (§2.2) and check the `dt.*` catalog before creating any custom metric — host, process, service, container, and consumption metrics almost certainly already exist for your question.
2. **Author everything new in DQL against `dt.*` keys.** Keep metric selectors only as the maintenance language for existing Data Explorer / Dashboards Classic assets, and migrate those with the built-in converter when you touch them.
3. **Design dimensions before first ingest.** Bounded values only — never request/user/session/trace IDs or timestamps. Do the tuple cross-product math (§7) while it is still a design review, not an incident. Put per-request detail in logs and spans, where it belongs.
4. **Namespace custom keys** with an org or app prefix (`acme.payments.*`). `dt.` is reserved and rejected. On Metrics Classic, count keys gain a `.count` suffix — expect it in metric selectors; Grail does not add it, so DQL queries use the key as sent.
5. **For OTLP, settle temporality at the source**: `OTEL_EXPORTER_OTLP_METRICS_TEMPORALITY_PREFERENCE=DELTA` in SDKs, or `cumulativetodelta` (with `max_staleness`) in the Collector — and confirm your histogram shape survives the type mapping (§4.4) before promising percentiles — explicit-bucket histograms stay histograms only with **Advanced OTLP metric dimensions** enabled.
6. **Attach metadata at ingest** (unit, display name, description — §2.4) so every downstream chart and Davis analysis renders meaningfully without tribal knowledge.
7. **Derive recurring aggregates from records early** via OpenPipeline extraction (§5) — it is forward-only, so every week of delay is a week of history the metric will never have. Keep the raw records for forensics per the FAQ-09 rule.
8. **Decide retention deliberately.** The included 15 months at full 1-minute resolution covers most operational analysis; extend the default metrics bucket (up to 10 years) only for genuine long-horizon needs, and remember Classic comparisons older than 14 days are already downsampled.
9. **Put the "Metric & Dimension Usage + Rejections" dashboard on your platform team's rotation** — cardinality rejections are silent by design, so make them visible by habit.

> **Sources:** the individual practices are grounded in the sections cited above (§2, §4, §5, §7, §8, §9). **Derived:** the ordering and emphasis are an authoring synthesis — the docs do not publish a consolidated metrics-design checklist.

11. Common Gotchas

Symptom	Cause	Fix
`timeseries avg(builtin:…)` fails to parse or returns nothing	Classic `builtin:` keys are not in Grail; the colon also reads as DQL parameter syntax	Query the Grail twin (`dt.host.cpu.usage`); backtick-quote any key containing special characters

Symptom	Cause	Fix
`percentile` / `median` in `timeseries` returns no rows (with a WARNING notification)	Missing `rollup:` — required for these functions on gauge/count metrics, except `dt.service.request.response_time` and `…service_mesh.response_time`	Add `rollup: avg` (or `min` / `max` / `sum`)
OTel counter never arrives	Cumulative temporality — Dynatrace ingests delta counters only	Set `…TEMPORALITY_PREFERENCE=DELTA` or add the Collector's `cumulativetodelta` processor
OTLP export fails outright	gRPC endpoint or JSON encoding — neither is supported	HTTP + binary protobuf (`http/protobuf`)
Histogram percentiles poorer than expected	Exponential OTLP histograms store only min/max/sum/count — buckets are dropped; OTLP summaries are unsupported	Use explicit-bucket histograms (Dynatrace 1.300+) **and** enable Advanced OTLP metric dimensions — without it, explicit-bucket histograms are ingested as counters
`metrics` command shows nothing older than ~10 days	Discovery command timeframe is limited to the last ten days	Use `timeseries` for historical values; `metrics` is for discovery only
Ingested data points vanish without error	Timestamp outside the acceptance window (−1 h to +10 min), or a new dimension tuple rejected after a cardinality limit	Fix producer clocks/batching; check the "Metric & Dimension Usage + Rejections" dashboard
Custom count metric key "changed" in metric selectors	Metrics Classic suffixes count keys with `.count`; Grail does not	In metric selectors, query the suffixed key; in DQL, the key as sent
New hosts/pods missing from an old custom metric, existing ones fine	Cardinality limit reached — new tuples rejected, existing series continue	Reduce dimensions; the limit evaluates a 30-day sliding window, so trimmed cardinality recovers on its own
OTLP metric arrives without a unit	Instrument unit exceeded 63 characters and was dropped	Shorten the unit string
Year-over-year Classic comparison looks "smoothed"	Classic downsampling: data older than 28 days is stored at 1-hour resolution (1-day beyond 400 days)	Use Grail (`dt.*` via DQL) — 1-minute granularity over the entire retention window
Extracted metric has no history before its creation date	Metric extraction is forward-only; records are not backfilled	Create extraction rules early; scan records for the historical portion
Duplicate dimension key on one ingest line	Protocol accepts only one value per key	Deduplicate keys in the producer

> **Sources:** each row is grounded in the sections above (§3, §4, §5, §6, §7, §8) and their cited pages; rows marked from live validation were confirmed on a tenant on 07/07/2026; the `rollup:` warning row was re-confirmed 10/02/2026.

> ⚠️ **Disclaimer:** This content is AI-generated, community-driven, and **not supported by Dynatrace**. Metric backends, OTLP ingest behavior, limits, and DPS billing rules evolve across releases — always verify current mechanics against the Dynatrace documentation and your own tenant before relying on a specific limit or cost claim.